

JavaScript Core Series

Module 01: Mastering Variables

Presented by: Your Repo Core
Data & Architecture Deep-Dive

What is a Variable?

A **variable** is a named container used to store data values in memory.

- **The Analogy:** Think of a variable as a labeled storage box.
 - The **label** is the identifier (`age`).
 - The **contents** inside are the value (`25`).
- **The Reality:** Instead of forcing your brain to remember raw hardware memory addresses like `0x7FFF5FBFFD8` , you write readable code.

Variables in Memory: The Engine Room

The JavaScript Engine (e.g., V8) manages variable storage using two specialized structural environments:

1. The Stack

- High-speed, execution-ready storage.
- Holds **primitive values** (Numbers, Strings, Booleans) and variable identifiers.

2. The Heap

- Large, unstructured memory space.
- Holds complex **reference types** (Objects, Arrays, Functions).

Lifecycle: Declaration vs. Initialization

Setting up a variable involves two distinct runtime behaviors:

- **Declaration:** Registering the variable identifier within its scope execution environment.
- **Initialization:** Allocating memory and assigning the baseline/initial data value.

```
let targetScore;           // Declaration (Defaults to 'undefined')
targetScore = 100;         // Initialization / Assignment

const maxLimit = 500;      // Inline Declaration & Initialization
```